



LexRAG: Retrieval-Augmented Legal Reasoning

Semantic Search & Hybrid Retrieval over the Caselaw Access Project

Motivation & Background

Problem

Legal reasoning is inherently precedent-driven. Practitioners must identify prior rulings that support or contradict a given legal claim — a task that is time-consuming, error-prone, and inaccessible at scale using traditional keyword search.

Objectives

Enable semantic search over 6.7M US case law records using dense vector embeddings, supplemented by keyword matching for improved coverage. Support claim-level retrieval — surfacing cases that explicitly contradict or corroborate a legal proposition. Deliver explainable answers with traceable reasoning paths via an instruction-tuned language model synthesis layer.

Dataset

common-pile/caselaw_access_project (HuggingFace). Digitised US federal and state court opinions spanning over three centuries, comprising ~6.7M full-text rulings with metadata including court, jurisdiction, and decision date. Since the dataset was too big for training, we used the first 15,000 rows.

Retriever

Hybrid Retrieval Strategy

Case text is chunked into semantic segments and encoded with a sentence encoder. Each chunk is stored in a PostgreSQL database alongside its dense embedding vector. At query time, dense vector search using cosine similarity produces a ranked list of relevant chunks, while BM25 sparse retrieval produces a second ranked list. These two ranked lists are fused using Reciprocal Rank Fusion to generate a unified ranking of candidate chunks. Finally, the top results are reranked using the cross-encoder ms-marco-MiniLM-L-6-v2 to improve relevance ordering.

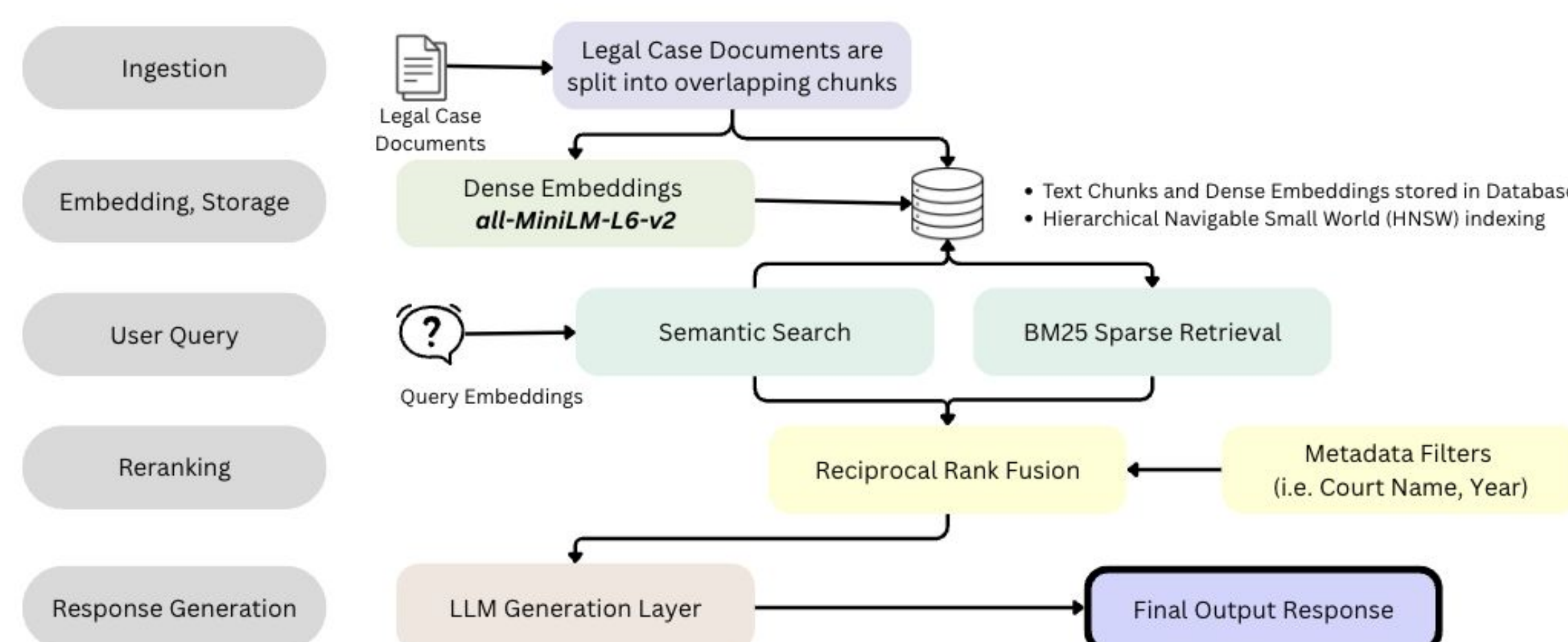


Fig. 1: Diagram of our RAG Pipeline

Response Generation

Prompt Engineering

We prompted the model to only use information in the retrieved documents, and to specify in its response if it is using outside knowledge. We also made sure that the model does not guess or infer missing facts. This way, we were able to limit the model hallucination, especially because the effects of misinformation would be very detrimental in a legal setting.

Results

Retrieval

Hybrid retrieval (dense + BM25 RRF) achieves Recall@10 = 0.70 and MRR = 0.77, as well as NDCG@10 = 0.64, outperforming sparse-only BM25 by +30pp on Recall@10. The dense component is most impactful for matching legal reasoning due to it being able to match multiple terms and specific phrases to the given query.

Answer Clarity

Given many types of example queries, the model responds in a clear and structured way, with direct reference to retrieved documents as well as detailed explanations for each aspect of the questions given by the user.

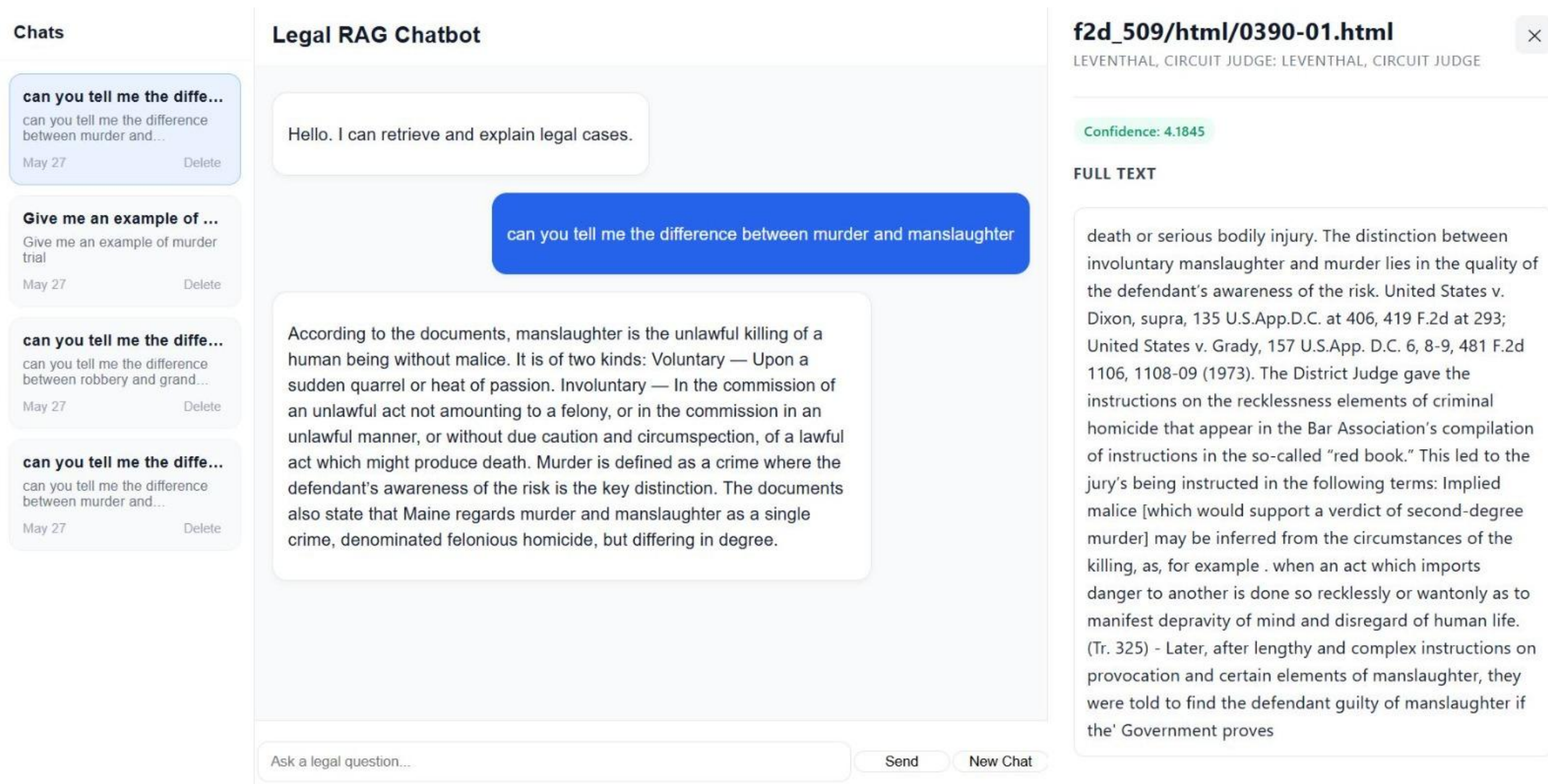


Fig.2: RAG Chatbot view – [Left] Chatbot where user can type questions. [Right] Expanded Reference

Model choice

The model we decided on is Gemma-3-4B-IT. We chose it to leverage a model that is already strong conversationally and responds well to the provided factual context, while still maintaining good performance independently of the hardware. We also tried a variety of different similarly sized models, but landed on Gemma because it showed the best results in sticking to the facts while still providing human-like responses.

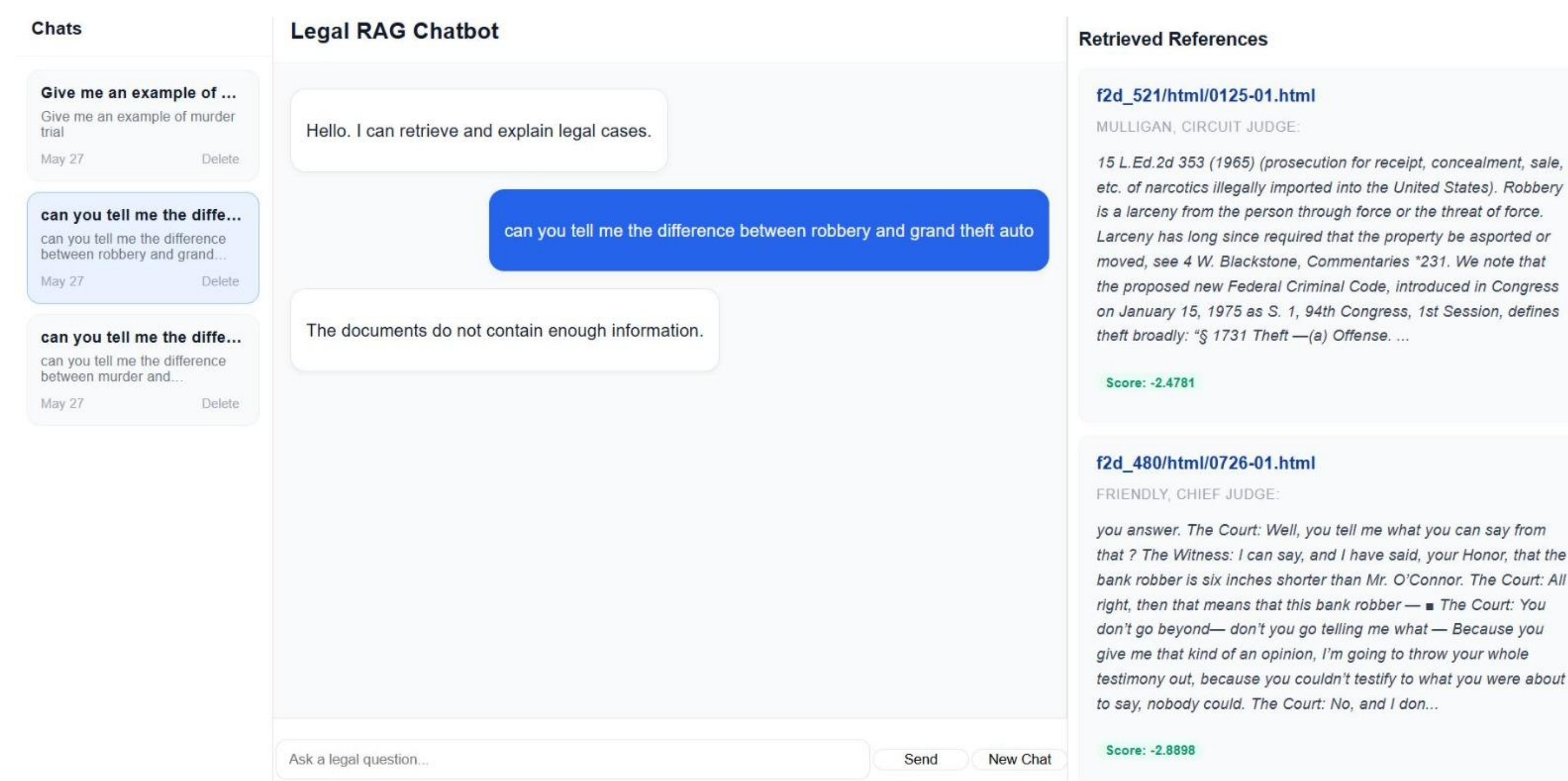


Fig. 3: Chatbot response when the retrieved documents are not able to answer the query